

15 Control and coordination

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A nervous impulse follows depolarisation when voltage-gated sodium channels open and repolarisation via potassium channels. Myelin permits saltatory conduction. Chemical synapses release neurotransmitter following calcium influx; transmission is one way because vesicles and receptor sites differ across the cleft. Hormones travel in blood and generally act more slowly but for longer.

Worked example: follow the reasoning

Blocking acetylcholinesterase prolongs acetylcholine in a neuromuscular synapse, so postsynaptic receptors are activated for longer.

Quick check · multiple choice

A Why can myelin increase impulse conduction speed?

[1]

- A It removes ion channels everywhere
- B It reverses impulse direction
- C Impulses regenerate at nodes
- D It makes synapses electrical

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain how myelination increases conduction speed and why chemical synapse transmission is directional.

[3]

Answers and explanation

Quick check: C. Impulses regenerate at nodes

Question 1 [3 marks]

Action potentials regenerate at nodes of Ranvier, reducing membrane to depolarise [2]; transmitter vesicles are presynaptic and receptor proteins postsynaptic [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

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